A3-PS033-TC354

DNS-Based Discovery of Diagnostic Services

Merge Conflicts
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Problem Description

Abstract
This project tests whether diagnostic services can be discovered through DNS without relying on hardcoded IP addresses or a central registry. A DNSSEC-signed lab zone will publish DNS-AID-style metadata for three mock services: a latency probe, a route-trace responder and a DNS inspector. A discovery client will query the zone, validate DNSSEC, check the metadata format and then make an authenticated request to the selected service.

Problem points
- Endpoints can move, making hardcoded addresses unreliable.
- DNSSEC can protect DNS data integrity, but does not grant permission to call a service.
- Metadata may be malformed, and a valid service may still be unavailable.
- The prototype will test these cases separately in a controlled lab.

Objective: demonstrate discovery, validation, authenticated invocation and clear failure handling using one signed test zone.

Background

DNS is already used to locate network resources, while DNSSEC adds origin authentication and integrity for signed DNS data. The DNS-AID draft explores publishing structured information that helps an agent discover a service and its capabilities. This project implements only a small, pinned subset of that proposal in a private test environment.

The prototype contains one DNSSEC-signed zone, a publisher for service metadata, one discovery client and three mock diagnostic services. The client must not treat a successful DNS lookup as permission to invoke a service. It checks the signature chain, parses the metadata, applies a separate authorization rule, and only then sends an authenticated request.

The lab will remain self-contained. It will not build an Internet-wide service registry or claim that DNSSEC proves a service is safe or trustworthy.

Literature & RFC Review

DNS-AID draft - provides the discovery model for advertising agent or service metadata through DNS. The prototype will document and pin the exact subset of fields it uses rather than assume full draft support.

DNSSEC - RFC 4033, RFC 4034 and RFC 4035 define DNSSEC concepts, record formats and validation behavior. They guide the signed test zone and client-side validation checks.

Authenticated invocation - DNS discovery and DNSSEC do not replace service-level authentication or authorization. The mock APIs will require a token or equivalent lab credential after discovery.

Design gap addressed - many simple discovery examples stop after returning an endpoint. This project also checks malformed metadata, bogus DNS data, denied invocation and unavailable services, reporting each as a distinct result.

Proposed Solution

The client follows six steps:
1. Discover: query the configured DNS name for the selected diagnostic capability.
2. Verify: validate the DNSSEC chain and reject bogus or unsigned data where signatures are required.
3. Parse: check that required metadata fields exist and have valid types and values.
4. Authorize: apply a separate local policy to decide whether this client may call the service.
5. Invoke: send an authenticated request to the endpoint from the validated metadata.
6. Report: return the result or a specific error without silently falling back to an untrusted endpoint.

Mock services: Latency Probe returns a simulated latency result; Route Trace returns a fixed lab route; DNS Inspector returns a controlled lookup summary. Each service exposes metadata in the pinned format and requires authentication.

Optimization Proposed

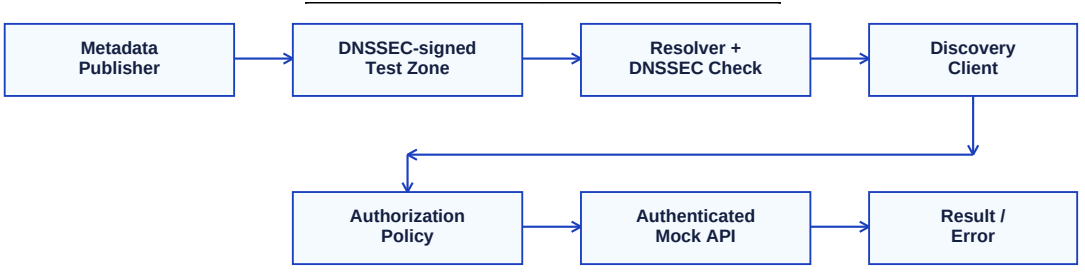
The main improvement is to make each trust decision explicit instead of treating discovery as a single successful/failed lookup.

Area | Basic approach | Proposed approach
Endpoint changes | Hardcoded address | Resolve from DNS metadata
DNS integrity | Accept answer | Require DNSSEC validation
Metadata | Use returned fields | Validate schema and required fields
Permission | Assume discovery is enough | Check service authorization separately
Failures | Generic error | Report bogus data, malformed metadata, denied access or unavailable service
Testing | Happy path only | Positive and negative test cases

Evaluation measures: successful discovery rate, correct rejection of invalid records, correct classification of failures, and successful authenticated calls to authorized mock services. No performance improvement will be claimed unless measured.

Solution Architecture & Design

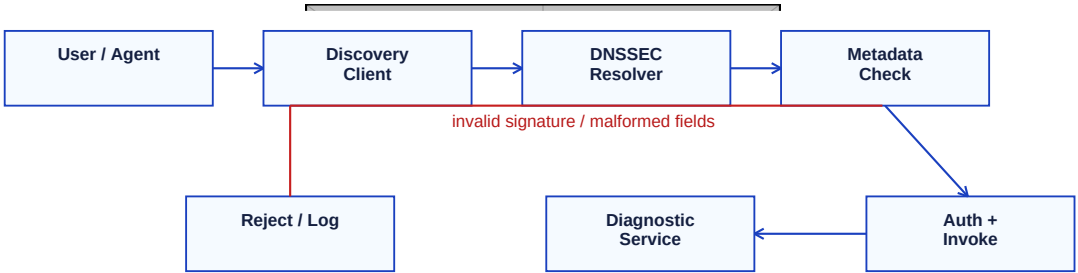
Discovery, metadata integrity, authorization and service execution are separate steps.



Security boundary: DNSSEC validates DNS data; the authorization policy independently decides whether the client may invoke the service.

Data Flow Diagram

Normal path and rejection path are both tested; invalid data must stop before invocation.



Expected output: a diagnostic result or a clear error. A DNSSEC-valid answer alone never grants invocation permission.

Deployment Timeline

Phase 1 - Setup: create the repository and pin the DNS-AID draft revision and metadata fields.

Phase 2 - DNS lab: configure one test zone, enable DNSSEC signing, and test valid and bogus answers.

Phase 3 - Build: implement the publisher, discovery client and three mock diagnostic services. Add schema checks and authenticated invocation.

Phase 4 - Test and demo: test valid discovery, bogus DNS data, malformed metadata, denied access and unavailable services. Record results and prepare a short pseudocode walkthrough.

Completion means every case is repeatable and the report separates DNS data integrity from service authorization and trust.

[1] IETF, "DNS-AID: AI Agent Discovery via DNS," draft-mozleywilliams-dnsop-dnsaid. Draft; pin the revision used by the prototype.

[2] R. Arends et al., "DNS Security Introduction and Requirements," RFC 4033, March 2005. https://www.rfc-editor.org/rfc/rfc4033

[3] R. Arends et al., "Resource Records for the DNS Security Extensions," RFC 4034, March 2005. https://www.rfc-editor.org/rfc/rfc4034

[4] R. Arends et al., "Protocol Modifications for the DNS Security Extensions," RFC 4035, March 2005. https://www.rfc-editor.org/rfc/rfc4035

[5] IETF, "Service Binding and Parameter Specification via the DNS (SVCB and HTTPS Resource Records)," RFC 9460, November 2023. https://www.rfc-editor.org/rfc/rfc9460

Implementation note: the exact DNS-AID record layout depends on the pinned draft revision. The prototype report should state the revision and the fields actually implemented.

Thank You for Your Attention!

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